5 IC Fabrication

IC FABRICATION

Basics of IC fabrication:

Step 1: Take a p-type substrate.

Step 2: Grow native oxide.

Pass the native oxide layer over the p-type substrate of silicon, hence resulting in the growth of silica.

Step 3: Photolithography

It is used for the patterning of the surface.

Step 3a): Apply the photoresist layer.

Step 3b): Patterning

- Expose the surface to the UV rays, and only those parts of the photoresist layer will be exposed to the rays which need to be removed.
- Here, a positive photoresist is used, which on the exposure to the UV rays, will dissolve in the developer's solution. And then that particular part will be removed.

Step 3c): In the development, the pattern is put over the photoresist layer.

Step 4: Etching

- Now, the transfer of the pattern on the silica is required.
- In this process, the surface of the wafer is exposed to hydrofluoric acid, which etches the SiO_2.
- The part of the silica covered with the photoresist layer is safe, whereas the part exposed to the hydrofluoric acid is etched out, opening a window of the substrate.

Step 5: Remove photoresist

- After the transfer of the pattern over the silica, the photoresist layer is no more required. A window is opened, and the substrate is exposed to the external environment.

Step 6: n-type diffusion

- This process includes the passing of n-type gas over the surface. It will diffuse because the concentration of n-type is high at the top and low at the bottom.

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- There are two types of photoresists a) positive and b) negative.
- For a positive photoresist, on exposure to the UV rays, it will dissolve. Whereas for negative photoresists, the part which is not exposed to UV rays will not dissolve.

Rack Your Brain

Why is silicon used in the manufacturing of electronic devices and not germanium?

- The n-type layer will be formed where the substrate is exposed to the n–type gas, and the SiO_2 layer acts as a mask. Hence no diffusion takes place there.

Step 7: Metallisation

- After forming the p–n junction, contacts are to be provided.

Step 7a): Pass the oxide gas onto the surface so that the SiO_2 layer is formed.

Step 7b): Apply the photoresist layer for patterning.

Step 7c): Then do the masking and pass the UV rays onto the part which needs to be removed.

Step 7d): Put the substrate into the developer solution.

Step 7e): Etching with hydrofluoric acid is to be done next.

- Etching of aluminium metal is done by using an $H_3PO_4 + HNO_3$ solution.
- Mask those areas where we want the layer.

Step 7f): Remove the photoresist layer.

Step 7g): Aluminium deposit

- Cover the top layer with Aluminium (Al).
- But due to this, the wafer will convert into a short circuit. Hence, patterning has to be done.

Step7h): Patterning (Photolithography +Etching)

- Removal of extra metal from the top layer is done. Etching of Aluminium is done using the H_3PO_4 + HNO_3 solution.
- As a result, we get 2 contacts, one for p-type and the other for n-type forming the p-n junction diode.

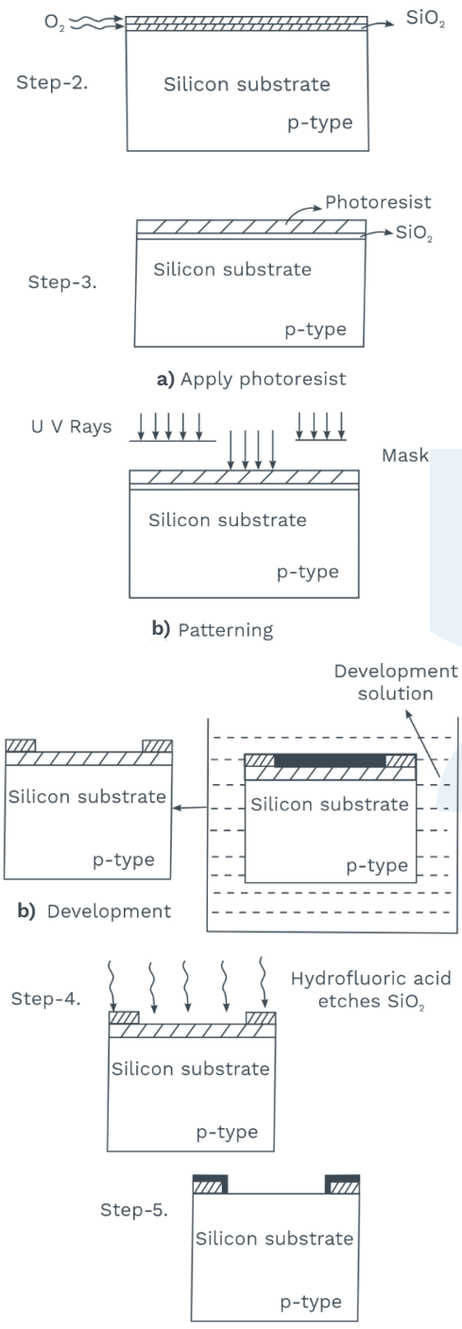
Step-1.





Rack Your Brain

What is the name of the transistor formed when the gate terminal is isolated by SiO_2?



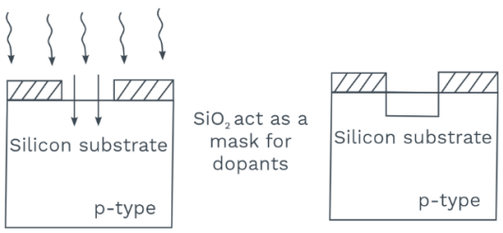
Rack Your Brain

What is the main purpose of metallisation?

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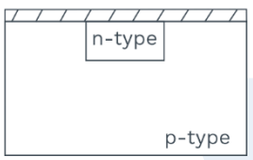
The silica layer provides insulation to the surface and has the advantage of passivisation against the dopants.

Step-6.

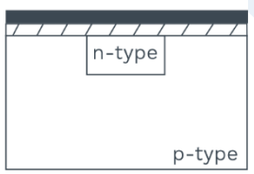


Step 7:

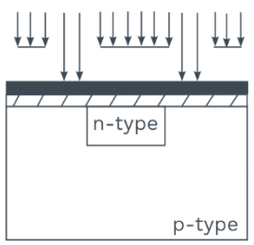
7.a) Oxidize



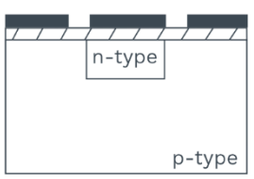
7.b) Apply Photoresist



7.c) Pass U V Rays



7.d) Development



Previous Years' Question

A silicon wafer has 100 nm of oxide on it and is inserted in a furnace at a temperature above 1000°C for further oxidation in dry oxygen. The oxidation rate:

a) Is independent of current oxide thickness and temperature.

b) Is independent of current oxide thickness but depends on the temperature.

c) Sl down as the oxide grows.

d) Is zero as the existing oxide prevents further oxidation.

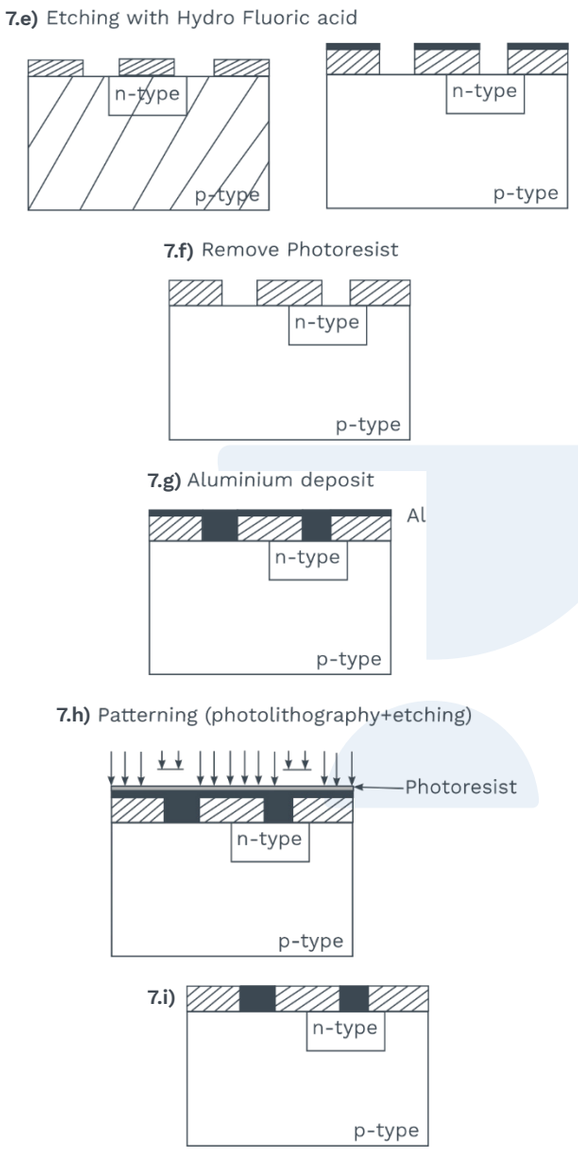


Fig. 5.1 Basic Steps Involved in IC Fabrication

Oxidation:

- Oxidation is basically the growth of oxide.
- The process of oxidation is used to provide a mask against the dopants.
- The growth of silica over a silicon surface is very easy. Just bypassing the oxygen gas onto the layer, we can initiate the chemical reaction

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Which form of oxygen is used in the wet oxidation process?

that happens automatically.

- This SiO_2 layer is a very good insulator and the dielectric property of this oxide layer makes it very beneficial in the fabrication of devices.

There are 2 types of oxidation:

1) Dry oxidation:

- When only oxygen gas is passed over the substrate.
- The oxide quality is very high.
- The dry oxidation has a lower growth rate than wet oxidation.
- Used in the growth of thin oxides like pad oxide and gate oxide.
- It has a high density of oxide layer, hence high breakdown voltage.
- Chemical equation involved in this process-

$$Si + O_2 \xrightarrow{1100°C} SiO_2$$

2) Wet oxidation:

- In this process, water is used instead of oxygen.
- This process has a higher oxidation rate and is also used to form thick oxides.
- Chemical equation involved in this process:

$$Si + 2H_2O \xrightarrow{900–1000°C} SiO_2 + 2H_2$$

- Hydrogen (H_2) is a by-product of this process. Assume a silicon substrate having thickness 'x μm' given in the figure.

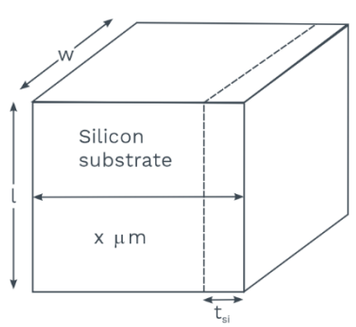


Fig. 5.2 Silicon Substrate Having

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Wet oxidation is a form of hydrothermal treatment as it is the oxidation of dissolved components in water using oxidation.

Grey Matter Alert!

The key aspects of each technology involved in etching are the safety of the operators, cost-efficiency, and control of dimensions.

Thickness 'x μm' and Width 'W'

- After oxidation, the SiO_2 layer will be formed on the surface of the silicon substrate. This layer will consume some volume of silicon also.
- Hence, if the thickness of the silicon substrate is 'xμm', then after the oxidation, this width will reduce to "x'μm".

$$x - x' \neq t_{ox}$$

where

t_{ox} is the width of the silica

- Above equation follows because if the thickness of the SiO_2 layer formed is t_{ox}, it will consume a part of the silicon also having width t_{si}.

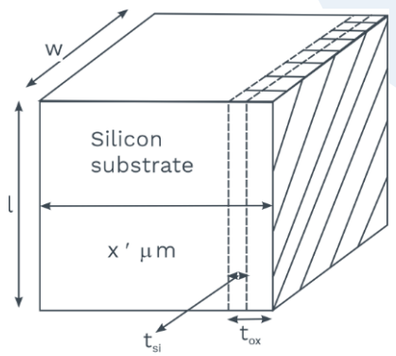


Fig. 5.3 Silicon Substrate After Oxidation

- Volume of SiO_2 formed = $l \times W \times t_{ox}$
- Volume of Si consumed = $l \times W \times t_{si}$

Let N_{si} be the number of silicon atoms per unit volume:

$$N_{si} = 5 \times 10^{23}/cm^3$$

Let N_{ox} be the number of silicon diode molecules per unit volume:

$$N_{ox} = 2.3 \times 10^{22}/cm^3$$

Now,

For 1 SiO_2 molecule $\rightarrow$ 1 Silicon atom is consumed

If the number of SiO_2 molecules is $N_{ox} \times l \times W \times t_{ox}$, then the number of silicon atoms consumed is

Definitions

Isotropic: When etching can be done in a directional way, the only selective part gets etched due to this property.

Example:

If the ions hit vertically, the etching will happen only in the vertical direction.

equal to $N_{si} \times l \times W \times t_{si}$

$$N_{ox} \times l \times W \times t_{ox} = N_{si} \times l \times W \times t_{si}$$

$$\text{Volume of silicon atoms consumed} = \frac{N_{ox}}{N_{si}} \times \text{Volume of } SiO_2 \text{ formed}$$

$$= \frac{2.3 \times 10^{22}}{5 \times 10^{23}} \times \text{Volume of } SiO_2 \text{ formed}$$

$$\text{Volume of silicon atoms consumed} = (0.46)\left[\text{Volume of } SiO_2 \text{ formed}\right]$$

Example 1:

A silicon wafer thickness of 100μm is oxidised to form SiO_2 of 10μm thickness. Find the value of x in the figure.

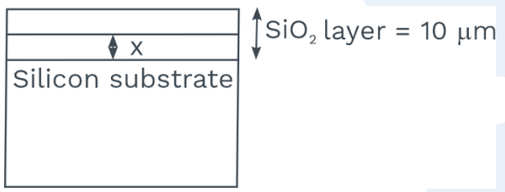


Fig. 5.4 Silicon Substrate

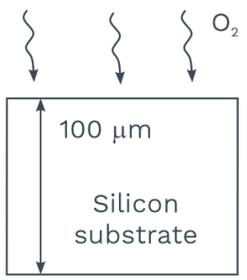


Fig. 5.5 Silicon Substrate After Oxidation

Solution:

Using the relation:

The volume of silicon atoms consumed is equal to (0.46) times the volume of SiO_2 formed.

Hence, x =(0.46) (10μm) = 4.6 μm

ETCHING

- This process is basically the removal of unwanted material in silicon integrated circuits.

Types of etching:

1) Wet chemical etching:

Steps involved in the wet chemical etching are:

- Transport of etchant to the surface
- Surface Reaction
- Removal of reaction products

Step 1: Transport of etchant to the surface:

- Put the whole wafer inside the etchant.

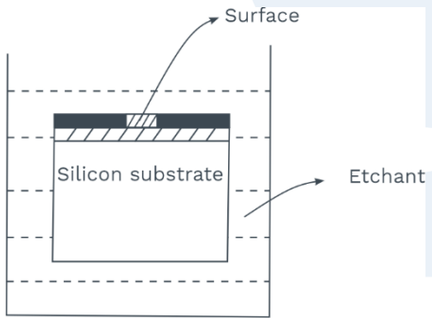


Fig. 5.6 Step 1 Involved in Wet Chemical Etching

- After the formation of the photoresist pattern, it can be used as a mask to etch the material present underneath.

Step 2: Surface Reaction:

- Concentration of the etchant is high at the top than that at the bottom. Hence, when the etchant tries to diffuse inside the silicon dioxide layer, it reacts with SiO_2.
- Usually, the etchant is hydrofluoric acid.

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Problems in wet chemical etching:

1) It is isotropic.

2) Maintaining the quality of chemicals is a big issue.

3) It is sensitive to impurities.

Definitions

Ion bombardment- The strong the negative field in the chamber of dry chemical, etching attracts the positive gas ions, and they strike over the silicon wafer with very high energy.

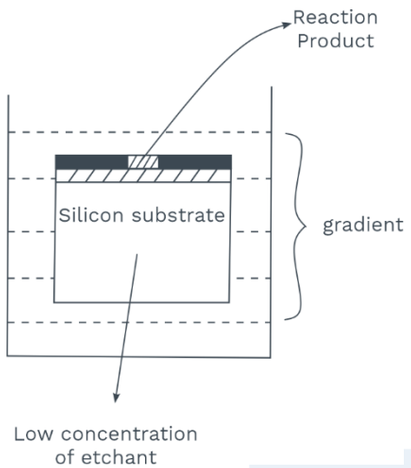


Fig. 5.7 Step 2 Involved in Wet Chemical Etching

Step 3: Removal of reaction products

- After the removal of the reaction product, only the part that is to be etched reacts with the etchant resulting in the required formation.

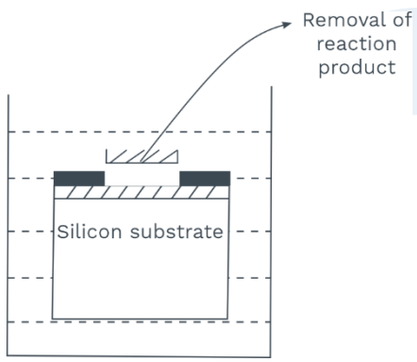


Fig. 5.8 Step 3 Involved in Wet Chemical Etching

2) Dry etching:

- The dry etching process is also known as plasma etching or reactive ion etching.
- In this process, the substrate emerges in a reactive gas (plasma). The layer to be etched (silicon/substrate) is removed by ion bombardment.
- The reaction products are carried away by the gas stream.

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Dry etching is also known as sputter etching because sputter refers to the knocking off the layer atoms by the gas ions inside the chamber.

Grey Matter Alert!

The gas strikes only vertically; hence the process is anisotropic, which is the main advantage of dry etching.

- It has anisotropic nature.

Steps involved in the dry chemical etching are:

- Multiple silicon wafers are placed over the wafer platter inside a gas chamber.
- The gas chamber is connected to a strong RF field (up to 13.56 MHz). This electric field ionises the gas inside the gas chamber.
- This gas is known as plasma.
- The chamber is connected to a strong oscillating field. So, in each cycle of the field, the electrons are electrically accelerated up and down inside the chamber.
- As the gas ions are heavy, the movement is very slow, whereas the electrons are lighter; hence they accelerate up and down faster.
- The electrons get deposited on the wafer, creating a very strong negative field on that surface.
- The gas ions, because of ion bombardment, can react chemically with the material of the surface of the samples.
- They can also transfer their energy to the atoms on the surface; hence those atoms knock off from the substrate material.

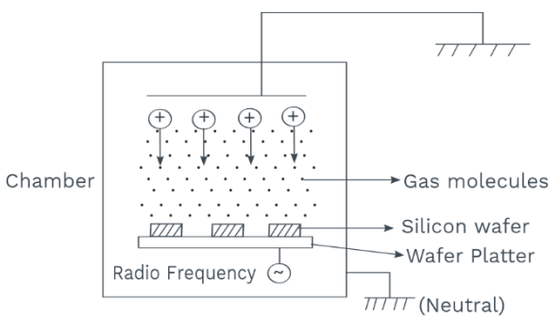


Fig. 5.9 Dry Chemical Etching Setup

Diffusion:

- This is a natural phenomenon
- Diffusion is the movement of material from high concentration to low concentration.

Types of diffusion:

Definitions

Solid solubility: Amount of material which is soluble inside the solid.

1) Infinite source diffusion:

Step 1: The source is permanently present in this process. The n-type carriers move towards the low concentration from the high concentration.

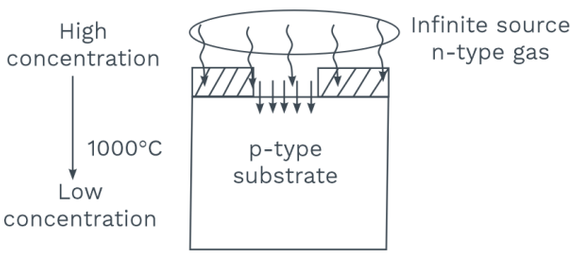


Fig. 5.10 Step 1 of Infinite Source Diffusion

Step 2: This diffusion of atoms occurs at a very high temperature. It is done to introduce dopants into the semiconductor.

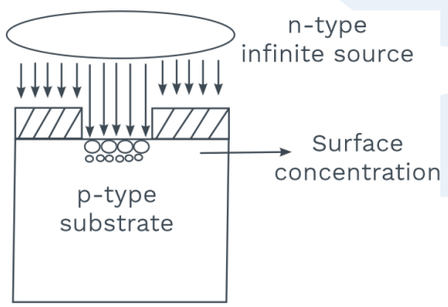


Fig. 5.11 Step 2 of Infinite Source Diffusion

- Surface concentration is constant because of the presence of an infinite source. And the surface concentration is equal to solid solubility.

Step 3: The diffusing of the n-type material from the infinite source continues till the point when the concentration of dopants is equal to the concentration of impurity atoms (acceptor ions) present in the substrate.

- That particular point determines the depth till which the definition will occur.

Definitions

DOSE is the number of atoms present per unit area.

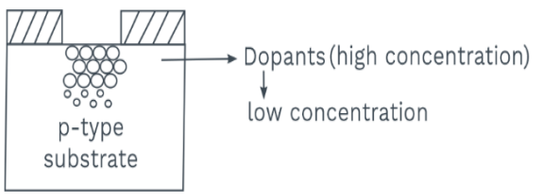


Fig. 5.12 Step 3 of Infinite Source Diffusion

2) Limited source diffusion:

- Source is present for some time before it is removed.

The conditions for this process are:

- **Predisposition:** The source is present, and diffusion occurs from higher to lower concentration.

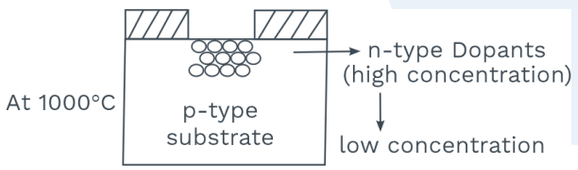


Fig. 5.13 Predisposition (Starting Stage)

- After the source is removed, the diffusion occurs deeper as in figure-

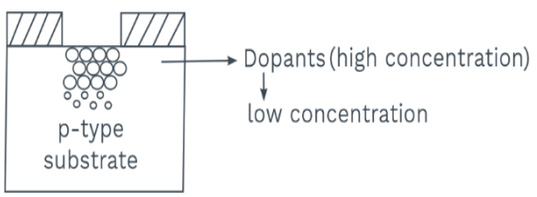


Fig. 5.14 Predisposition (Second Stage)

- **Drive-in:**

The stage in the figure below is the drive-in stage, where the molecules will rearrange themselves.

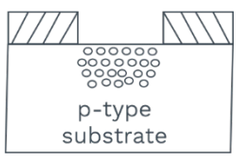


Fig. 5.15 Drive-in Stage

Grey Matter Alert!

Disadvantage of LOCOS:
The effective area for the fabrication of transistors gets very limited. For high performance, we must grow as higher several transistors as we can.

- The surface concentration is not constant as it decreases after heating.
- The number of dopant atoms is constant, i.e. the DOSE is constant.
- Because the number of atoms that have been diffused from the source will always be the same, but the depth will increase further.

ISOLATION TECHNIQUES

1) LOCOS [Local oxidation of silicon]:

- After the fabrication of a circuit, there are thousands of transistors present on a single chip. Hence, to separate those transistors to avoid the short-circuiting effect, isolation techniques are required.

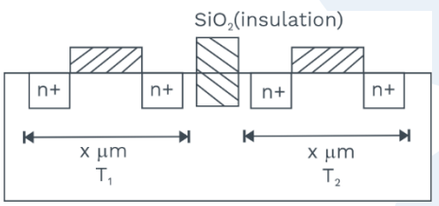


Fig. 5.16 Basic Setup After Fabrication of the Circuit

Step 1:
Take a substrate.

Step 2:
Assume taking a p-type substrate and then growing pad oxide onto the substrate.

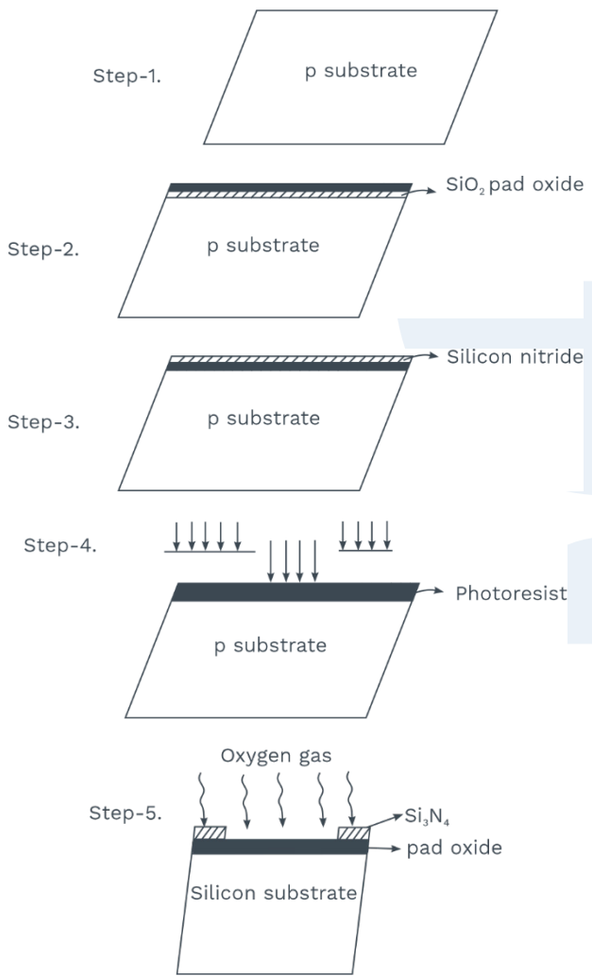
Step 3:
Deposit silicon nitride over the pad oxide.

Step 4: Patterning of Si_3N_4:
The patterning of silicon nitride is done using the following processes:

- First masking, then exposing with the UV rays and after that, development.
- Photoresist layer will be removed and Si_3N_4 is grown on the top layer wherever required because of etching.

Step 5: Passing oxygen gas

The part of the top layer exposed to the external environment reacts with the oxygen gas and hence silicon substrate to form silicon dioxide.



Result:

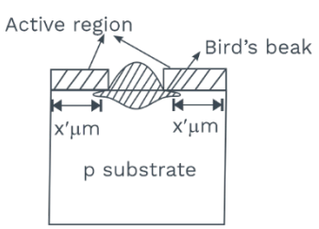


Fig. 5.17 Steps for LOCOS Technique

Grey Matter Alert!

Dishing effect is the difference between the height of the oxide in the spaces and the height of the metal in the trenches. Dummy fill techniques can reduce this effect.

Grey Matter Alert!

The difference in the shallow trench isolation technique is that the transistor's active the region is predefined, i.e. where we need to grow the transistor. In the remaining area, SiO_2 is deposited. By using this process, the bird's beak problem is avoided.

- The transistor can be fabricated in the active region, but the problem is the encroachment of the silicon dioxide over the active region, and hence it decreases the active area of the transistor.

This issue is also known as the bird's beak problem as the shape resembles the bird's beak.

2) Shallow trench isolation:

Step 1:

- After the patterning of silicon nitride (Si_3N_4) using photolithography and etching processes, the trench is formed in which silicon dioxide has to be deposited.
- Silicon nitride will act as a protective layer against the oxide gas.
- Dimensions are predefined, where we want to form a trench & the two transistors.

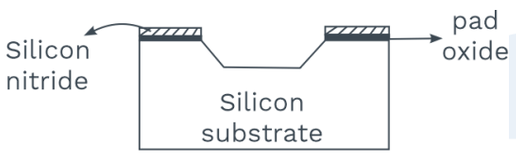


Fig. 5.18 Step 1 in the Shallow Trench Isolation

Step 2:

- Pass the oxygen gas, it will consume the silicon surface and form silicon dioxide.
- But to avoid this, a very thin layer of silicon dioxide is deposited over the trench surface so that the actual deposition of silicon dioxide occurs on the entire surface.

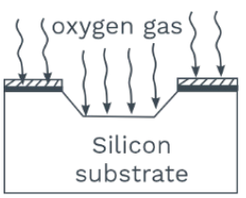


Fig. 5.19 a)

Definitions

Metal fill insertion: At the time of etching, the use of chemicals leads to chemical metal loss. After that reaction, the metal fills are inserted.

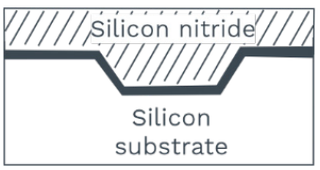


Fig. 5.19 b)

Fig. 5.19 Step 2 in the Shallow Trench Isolation

Step 3:

- Again, for the patterning of SiO_2, photolithography and etching processes are performed to eliminate. The unwanted parts of this silicon dioxide layer.

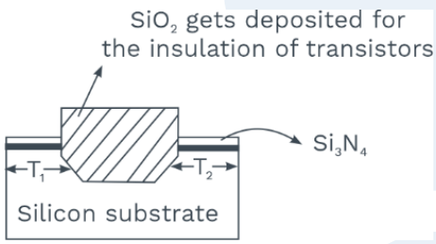


Fig. 5.20 Step 3 in the Shallow Trench Isolation

Step 4:

- Remove the silicon nitride layer as it is no more desired. The main reason for using the Si_3N_4 was to protect the silicon dioxide from being deposited in the unwanted areas.

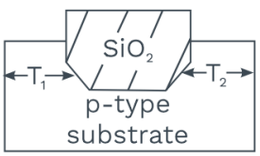


Fig. 5.21 Substrate after the Shallow Trench Isolation

Ion Implantation:

- This process is an alternative to the diffusion technique as diffusion consists of many drawbacks.
- It is not a natural phenomenon.
- Here, the ions with high energy are implanted.

Grey Matter Alert!

The chamber has to be connected to the ground because the electrons, while accelerating, can hit the walls of the chamber also.

- This technique is operated at room temperature. No prerequisite temperature conditions as in the diffusion are required in this process.

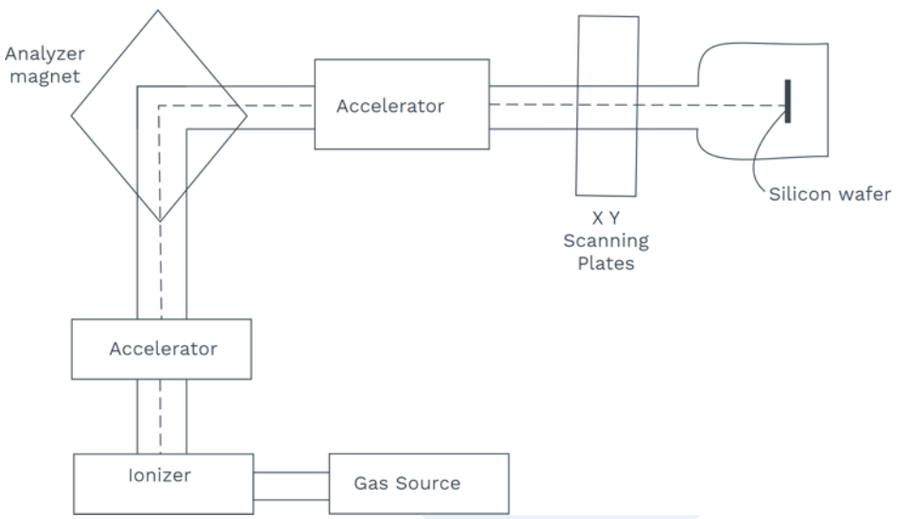


Fig. 5.22 Ion Implantation Set-up

- Gas source provides the required material (dopant gas) for implantation on the silicon wafer.
- Ioniser creates an electric field to collect the electrons; it ionises the gas for the future process.
- This ionised gas is passed onto the accelerator and the accelerate the ions coming from the ioniser.
- Analyser magnet (magnetic separator) separates the impurities present in the ionised gas.
- Again, the required energy is provided to the ions by passing them through the accelerator.
- The x–y scanning plates will make the ionised gas strike horizontally and vertically over the whole surface of the wafer.

Rack Your Brain

How can we avoid ion channelling?

Analyser magnet (Magnetic separator):

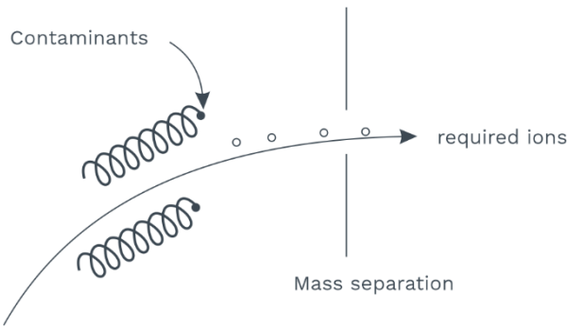


Fig. 5.23 Analyzer Magnet

- It depends upon the angle of deflection of every ion present inside the gas.
- Every ion deflects at a different angle; depending on the mass.
- Hence, the width of the slit is fixed, such as only the required gas with a particular mass of each ion will only be able to pass through it.

Photolithography:

- There are three types of lithography (depending upon the medium used):
- Optical lithography
- Electron beam lithography
- X-Ray lithography

Steps involved in the transfer of patterns:

Step 1: Application of a photoresist layer over the substrate.

Step 2: Creation of mask:

- And after passing light through it, the chemical properties of the photoresist will change. For a positive photoresist, the part exposed to light becomes soluble in the developer's solution. But the part of the surface protected by the mask does not dissolve.

Previous Years' Question

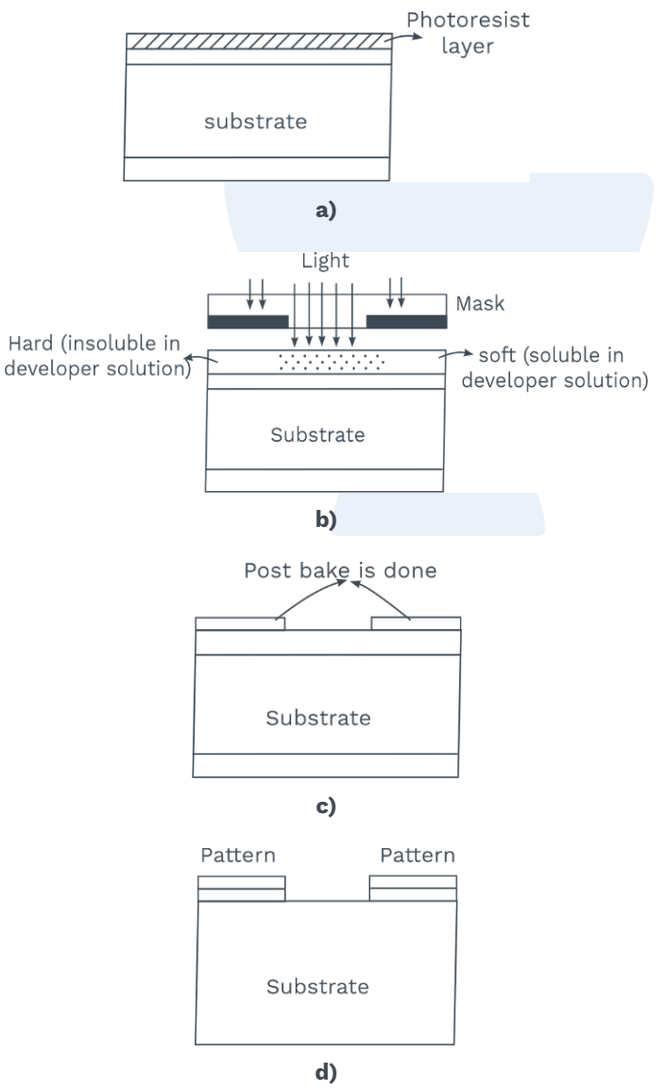
Thin gate oxide in a CMOS process is preferably grown using
a) wet oxidation
b) dry oxidation
c) epitaxial deposition
d) ion implantation

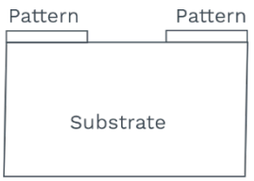
Step 3: Pre-Bake and Post-Bake processes.

- Pre-bake is done before the development process. In this, the moisture is removed after the application of the photoresist layer.
- Post-bake is done to harden the remaining photo-resist layer.

Step 4: Etching of silicon dioxide (using hydrofluoric acid) is done to etch away the unwanted part.

Step 5: The photoresist layer is also removed to get the required pattern.





e)

Fig. 5.24 Steps Involved in Transfer of Patterns

Flow diagram of the photolithography process:

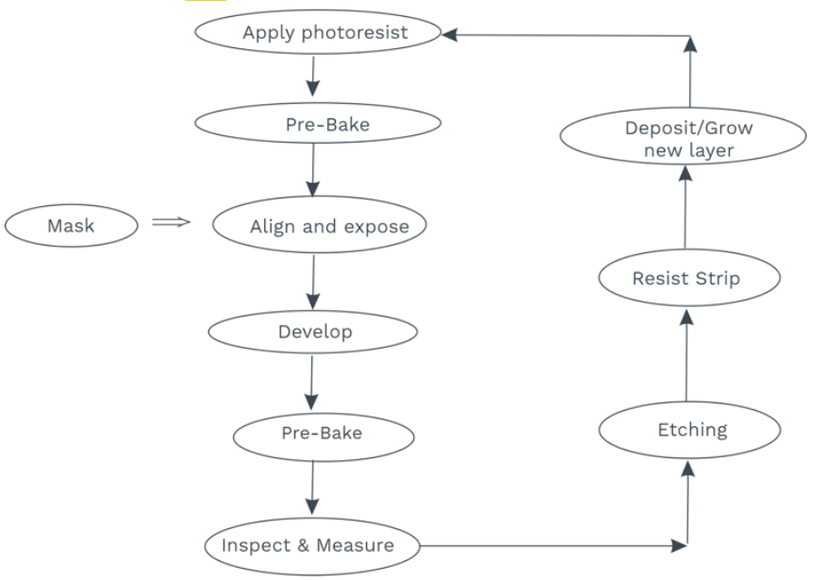


Fig. 5.25 Flow Diagram of Photolithography Process

Transfer of patterns:

1) Contact Printing

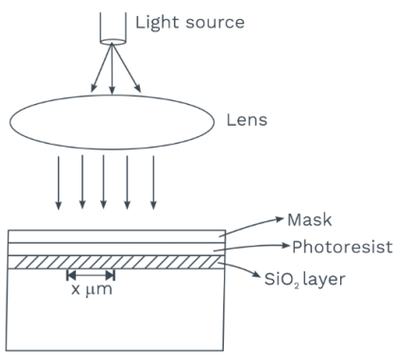


Fig. 5.26 Contact Printing Setup

Definitions

Diffraction: The phenomenon of the light that involves the bending of light at the edges is known as diffraction

- Minimum feature length is possible in this type, so the resolution is high.
- Mask durability is low, and the chances of wear and tear of the mask increase.
- Simplest method to transfer the patterns.
- The mask is in contact with the photoresist layer, so there is no chance of misalignment and only that part is exposed, which is required.

Masks: According to the type of material and the rays to which the material is exposed:

- Optical masks and X-ray masks
- Chromium absorbs the photo rays.
- Gold absorbs the X-rays.

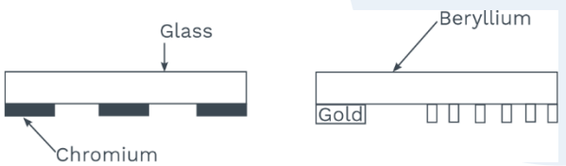


Fig. 5.27 Different Types of Mask

2) Proximity printing

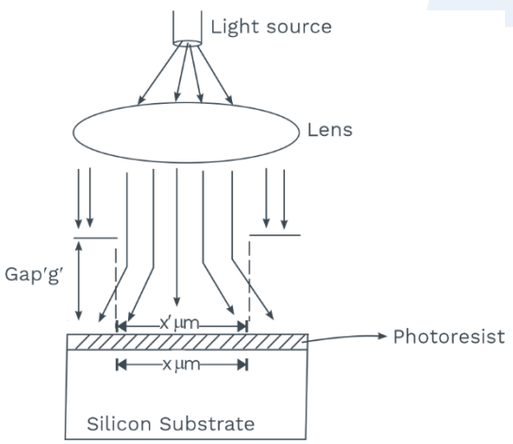


Fig. 5.28 Proximity Printing Setup

- The distance between the mask and the photoresist layer in this technique is increased, by which the life of the mask will also increase.

Definitions

Resolution: Number of pixels a screen can display within a given area.

Rack Your Brain

Why do we use X-rays instead of UV rays in projection printing?

Grey Matter Alert!

In Rutherford's model, gold absorbs the X-rays, from where the concept of an X-ray mask is taken.

- The diffraction occurs between the lens and the photoresist, and hence the resolution also decreases, which means that the designer will not get the minimum feature length.
- The area was required to be exposed till 'x μm' penetrates up to a longer length i.e. up to "x'μm" because of the bending of light rays.

3) Projection printing

- The most popular and usually used method for printing.

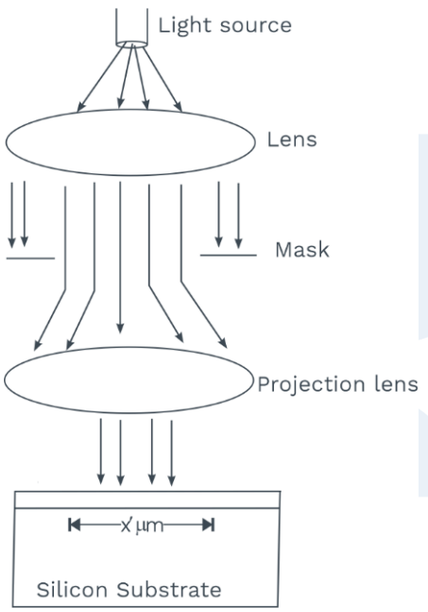


Fig. 5.29 Projection Printing Setup

- To avoid the problems faced in the previous two printing types, one extra converging lens that can focus the light is required.
- The projection lens will focus the light exactly on those parts required to be exposed.
- High durability of the mask and minimum feature length is achieved using this technique.
- The main drawback of this technique is the higher cost of the set up because instead of one, two lenses are used.

Minimum feature-length:

$$W_{min} \propto \sqrt{\lambda g}$$

Grey Matter Alert!

The word 'monolithic' is a Greek word which means 'one stone'. All the components are part of one chip hence this type of ICs are called monolithic ICs.

- It depends upon the wavelength of the light (λ) and the gap (g) between the mask and the photoresist layer.

The minimum feature length can also be written as:

$$W_{min} = \frac{K_1 \times \lambda}{NA}$$

Where

NA is the numerical aperture of the lens

λ is the wavelength of light

K_1 is a constant

Integrated circuits:

- An integrated circuit (IC) consists of transistors, diodes, resistors etc. and the interconnections of several components in a single small package.
- The size of an IC is very small.
- Various components are part of a small semi-conductor chip & these components cannot be removed.
- All the components are formed within the chip.

Advantages of the integrated circuit over discrete circuits:

1) Increased reliability.
2) Lesser weight and space requirements.
3) Low power requirements.
4) Low cost.
5) Ability to work at extreme values of temperature.

Disadvantages:

1) Components in an IC can't be replaced; hence if any one component fails, the whole IC has to be made new.
2) High-power ICs can't be produced.
3) Lack of flexibility in the IC's.
4) It is not possible to fabricate capacitances of higher value.

Four basic types of constructions for the manufacturing of IC's are:

1) Monolithic
2) Thin-film
3) Thick film
4) Hybrid

Monolithic and application-specific IC:

- Monolithic ICs are the most common type out of all.

Steps in the production of monolithic IC are:

Step 1:

A p-type silicon crystal is cut into thin wafers. One side of the wafer is polished to eliminate imperfections on the surface. This wafer is called the substrate.

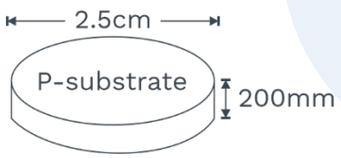


Fig. 5.30 Step 1 in the Production of Monolithic IC

Step 2:

The IC has to be produced on this wafer. These wafers are put into a diffusion furnace.

- A gas mixture of atoms of silicon and pentavalent atoms is passed over the wafer, forming a thin layer.

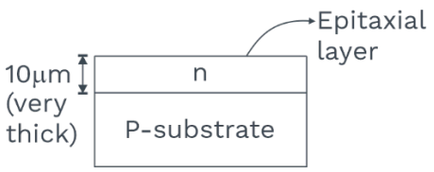


Fig. 5.31 Step 2 in the Production of Monolithic IC

Step 3:

To prevent the contamination of the epitaxial layer, a thin silicon dioxide layer is deposited over the entire surface by passing pure oxygen.

Step 4:

Diffusion is used to add appropriate materials onto the substrate and production of diodes, transistors, capacitors etc., are done.

Grey Matter Alert!

Scale of integration	Number of components
Small (SSI)	1 – 20
Medium (MSI)	20 – 100
Large (LSI)	100 – 1000
Very large (VLSI)	1000 – 10000
Super large (SLSI)	10000 – 100000

- The SiO_2 layer is etched before any impurity is added to the substrate.
- Because of etching, the epitaxial layer gets exposed and desired components are produced.

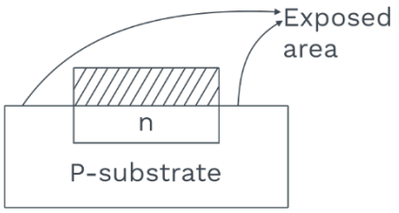


Fig. 5.32 Step 4 in the Production of Monolithic IC

Step 5:

Integrated circuits are produced at a low loss because they are produced in large numbers at once.

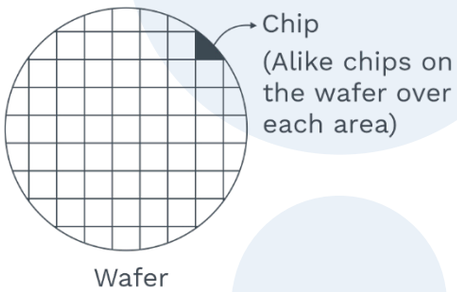


Fig. 5.33 Silicon Wafer for IC Production

Step 6:

The individual ICs are separated by a glass cutting like process after that; it is connected to the external leads. The ICs are then encapsulated to prevent contamination by the external environment.

Step 7:

Forming p-n-p junctions on the substrate:

- The wafer is put into a furnace, and the trivalent atoms are diffused into the epitaxial layer.
- The trivalent atoms change the exposed epitaxial layer from n-type to p-type semiconductor.
- An island of n-type material under the SiO_2 layer is formed.
- After this, pure oxygen is passed over the wafer to form a complete SiO_2 layer.
- A hole is then etched at the centre of this layer.
- Now, we pass trivalent atoms through this window form.
- The trivalent atoms diffuse into the epitaxial layer to form an island of p-type material, and a p-n junction diode is formed on the substrate.

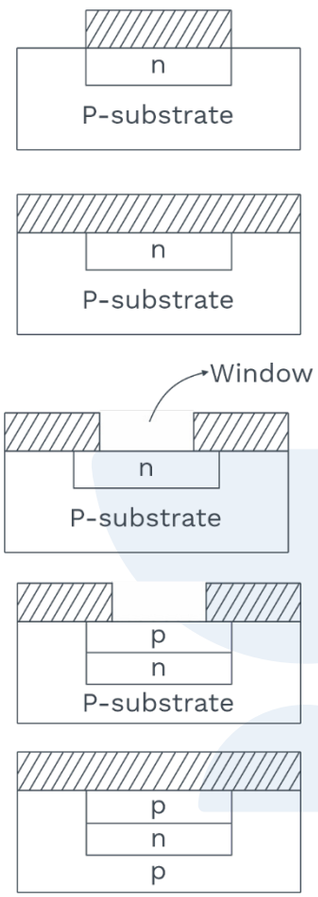


Fig. 5.34 Steps in Forming p-n-p Junctions on the Substrate

Step 8:

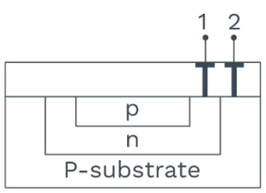
To attach the terminals for the manufacturing of application-specific ICs, SiO_2 layer is etched at the desired locations.

Step 9:

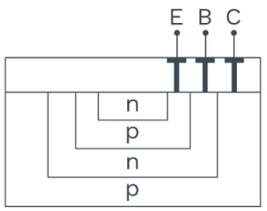
By depositing metal at the desired locations, electrical contact with the anode and cathode of the integrated diode are made.

The figures shown below depict the circuit of different electrical elements:

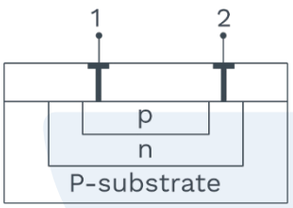
Diodes



Transistors



Resistors



Capacitors

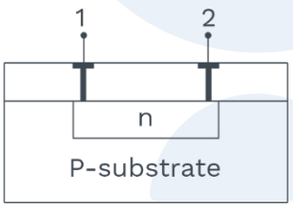


Fig. 5.35 Circuit of Different Electrical Elements on the Substrate

Chapter Summary



- The process of oxidation is used to provide a mask against the dopants.
- This SiO_2 layer is a perfect insulator, and the dielectric property of this oxide layer making it very beneficial in the fabrication of devices.
- There are 2 types of oxidation:
  1) Dry oxidation
  2) Wet oxidation

$$\text{The volume of silicon atoms consumed} = (0.46)\left[\text{Volume of } SiO_2 \text{ formed}\right]$$

- Etching is the removal of unwanted material in silicon integrated circuits.
- Types of diffusion-
  1) Infinite source diffusion: The source is permanently present in this process.
- Surface concentration is constant because of the presence of an infinite source
  2) Limited source diffusion:
- The source is present for some time before it is removed.
- After fabricating a circuit, there are thousands of transistors present on a single chip. Hence, to separate those transistors to avoid short-circuiting effects, isolation techniques are required.
- By using the shallow trench isolation process, the bird's beak problem is avoided.
- There are three types of lithography (depending upon the medium used):
  1) Optical lithography
  2) Electron beam lithography
  3) X-Ray lithography
- Analyser magnet (magnetic separator) separates the impurities present in the ionised gas.
- The minimum feature length can also be written as:

$$W_{min} = \frac{K_1 \times \lambda}{NA}$$

PRACTICE QUESTIONS

Q1 **Why does monolithic IC offer higher reliability than discrete component systems?**
a) Because of the closely matched electric elements in monolithic ICs.
b) Electric voltage is higher in discrete systems.
c) As there are fewer interconnections in the monolithic ICs.
d) All of the above

Q2 **In IC fabrication, dry oxidation, as compared to wet oxidation, produces:**
a) Inferior quality oxide
b) Superior quality oxide
c) Depends upon the concentration of oxygen gas
d) Depends upon the temperature used in the process

Q3 **In monolithic IC, the isolation technique used is/are:**
a) PN-junction
b) Dielectric junction
c) Beam-lead junction
d) none of the above

Q4 **In MOSFET fabrication, the insulating layer used is:**
a) Silicon dioxide
b) Silicon nitride
c) Aluminium oxide
d) Lithium oxide

Q5 **In an IC, the active components are:**
a) Resistors
b) Transistors
c) Diodes
d) Capacitors

Q6 **Metallisation is a technique which is used in IC for:**
a) Protection
b) Packaging
c) Isolation
d) Interconnection

Q7 **In an IC, the process involved in photolithography is:**
a) Photo etching
b) Making of a photographic mask
c) Both a) and b)
d) None of the above

Q8 **A silicon wafer with a thickness of 250 nm of oxide is placed in a furnace at a temperature of 1120°C for further oxidation in dry oxygen. Then the rate of oxidation:**
a) Slows down as the oxide rises.
b) Will be zero as the existing oxide will prevent further oxidation.
c) Will be independent of current oxide thickness but dependent on the temperature.
d) Will be independent of current oxide thickness and temperature.

Q9 **In an IC fabrication, silicon is considered because:**
a) It is more conductive material.
b) It is mechanically strong.
c) Silicon process characteristics which are best suited for IC design
d) All of the above

Q10 **In a monolithic integrated circuit, all the components are fabricated by a process is known as:**
a) Sputtering
b) Scattering
c) Diffusion
d) Oxidisation

Answer Key:

1) c)	**2) b)**	**3) a), b), c)**	**4) b)**	**5) b), c)**
6) d)	**7) c)**	**8) d)**	**9) c)**	**10) d)**